

AUSTIN MATTHEWS

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SUMMARY

Software Engineer specializing in full-stack development, AI/ML systems, and cloud-native applications. Shipped REDvsBLUE, a live cross-platform sports platform with 22 Clean Architecture modules on a Kubernetes-orchestrated backend. Achieved 0.64ms autonomous vehicle inference (16x faster than the 10ms industry standard). MS Computer Science, Seattle University (2026). Proven engineering leader with 3 years directing 26-person teams at Amazon, driving reliability from 94% to 99.7%.

EDUCATION

Master of Science in Computer Science | Seattle University

March 2026

Coursework: AI & Machine Learning, Distributed Systems, Software Engineering, Algorithms, Software Testing, Parallel Computing

Bachelor of Computer Applications | Himalayan University, India

Sep 2018

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Dart, Java, SQL, HTML/CSS

AI/ML: PyTorch, TensorFlow, OpenCV, scikit-learn, Transformers, FAISS, Ollama LLMs, LoRA/QLoRA/RLHF, Computer Vision, RL (PPO)

Mobile & Frontend: Flutter, React 19, Three.js, Tailwind CSS, Riverpod, Clean Architecture, Vite

Backend & Cloud: Supabase (PostgreSQL, RLS), FastAPI, Node.js, Express, Socket.IO, Firebase, Redis, AWS, GCP, Cloudflare Workers

Tools & Practices: Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Nginx, Sentry, Prometheus, Grafana, pytest, Agile/Scrum, TDD

PROJECTS

REDvsBLUE | Cross-Platform Sports Matchmaking App (Live)

Sep 2024 – Present

rvb-redvsblue.com | github.com/ausmatths/REDvsBLUE

- Shipped live cross-platform app (Android, iOS, web, macOS) with 22 Clean Architecture modules, 682 commits, and 99.9% uptime on a 12-replica Kubernetes cluster
- Engineered ELO/Glicko2-based matchmaking system with real-time queue management and multi-sport support (4 sports)
- Designed and deployed 30+ Firestore collections with 636 lines of security rules and 15+ optimized composite indexes
- Implemented CI/CD pipeline with GitHub Actions, Docker multi-architecture builds, and automated Firebase deployment
- Integrated payment processing (Stripe, PayPal, Google/Apple Pay) and real-time features (chat, notifications, tournaments)

Tech: Flutter, Dart, Firebase, Supabase (Auth, Firestore, Functions), Riverpod, GoRouter, GitHub Actions, GitLab, Cloudflare, Docker

GhostAI v2.0 | Enterprise AI Assistant Platform

Oct 2024 – Present

GitHub repository | github.com/ausmatths/GhostAI_v2

- Developed 3-tier multi-agent orchestration system with persistent memory achieving 99.9% uptime
- Optimized inference pipeline reducing response times by 73% with 70% memory reduction through efficient caching
- Achieved 12-120x performance improvements with comprehensive testing suite (95% code coverage) and 40+ Prometheus metrics
- Built LoRA/QLoRA/RLHF fine-tuning capabilities, multimodal AI, and CI/CD pipeline with automated deployments

Tech: Python, Streamlit, SQLite, Docker, Ollama LLMs, pytest, Prometheus, Grafana

Autonomous Vehicle Trajectory Prediction | AI/ML Research Project

Nov 2024 – Mar 2025

GitHub repository | github.com/ausmatths/real_vehicle_prediction

- Achieved 0.64ms inference time (16x faster than 10ms automotive requirement) with 1,601 inferences/second throughput
- Implemented multi-sensor fusion architecture (LiDAR, camera, radar) achieving 0.119m prediction accuracy
- Designed reinforcement learning system with PPO algorithm; validated across 9 adversarial scenarios with 100% pass rate

Tech: Python, PyTorch, TensorFlow, OpenCV, NumPy, Reinforcement Learning

EXPERIENCE

Graduate Assistant, Online Education Design | Seattle University

Jun 2024 – Jan 2026

- Developed interactive e-learning courses using Articulate 360 with SCORM compliance, deployed to enterprise LMS platforms for 10,000+ users
- Managed production website (seattleu.edu) serving 10,000+ daily users via T4 Site Manager CMS
- Implemented automated testing workflows reducing content bugs by 75% before deployment

AI Engineering Intern | RadicalX

Oct 2023 – Jan 2024

- Built ML-powered fraud detection models achieving 98.3% accuracy with 95% test coverage
- Developed LLM validation framework reducing hallucination errors by 67% in production systems
- Created real-time monitoring dashboard tracking 15 KPIs for model performance and data quality

NOC-Transportation Specialist (Team Lead) | Amazon India

Nov 2019 – Aug 2022

- Led team of 26 engineers, improving system reliability from 94% to 99.7% through proactive monitoring using internal dashboards and alerting tools
- Built automated reporting dashboards using SQL and internal tooling, reducing manual analysis time by 40% and errors by 85%
- Coordinated cross-functional incident response reducing MTTR by 35% during peak operations